

PANCREATIC DISORDERS — Clinical One-Page Review

Acute vs Chronic vs Autoimmune Pancreatitis vs Pancreatic Cancer. Bold = high-yield. [bracketed] = standard exam knowledge added beyond lecture.

	ACUTE PANCREATITIS	CHRONIC PANCREATITIS	AUTOIMMUNE PANCREATITIS	PANCREATIC CANCER
Definition / Key fact	Premature intracellular trypsinogen activation -> proteases digest pancreas + surrounding tissue. NFκB activation -> inflammation. Severity = balance proteolytic enzymes vs antiproteolytic factors (trypsin inhibitor, β2-macroglobulin, α1-antitrypsin, C1-esterase inh).	Chronic inflammation -> fibrosis + destruction of exocrine tissue . DM in advanced disease (islets involved).	Form of chronic pancreatitis that mimics cancer but responds to steroids . No acute attacks.	90% adenocarcinoma from pancreatic ducts; early local invasion + nodal mets. Neuroendocrine tumours = slower, better prognosis.
Epidemiology	3% of admitted abd pain; 2-28/100,000, rising. Overall mortality 10%. 80% mild (good outcome); 98% deaths in the 20% severe – 1/3 in 1st wk (multi-organ failure), rest from sepsis/infected necrosis.	Middle-aged alcoholic men. Steatorrhea once >90% exocrine tissue destroyed.	May occur alone or with Sjögren's, PSC, IBD.	10-15/100,000, rising to 100/100,000 if >70. Men 2x women. Most advanced at presentation. Overall survival 3-5%; median 6-10 mo (locally advanced), 3-5 mo (mets).
Causes / Risk	Gallstones, alcohol, pancreatic duct obstruction. (also drugs, ERCP, hyperCa, hyperTG – std).	~80% alcohol . Poor countries: calcific in non-alcoholics (malnutrition, trace element/micronutrient deficiency, cassava).	Autoimmune; ↑IgG4. Associated autoimmune disorders.	Age, smoking , chronic pancreatitis. 5-10% genetic: hereditary pancreatitis, HNPCC, FAMMM.
Clinical features	Severe constant upper abd pain , builds 15-60 min, radiates to back . N/V. Marked epigastric tenderness; guarding/rebound ABSENT early (retroperitoneal). Quiet/absent bowel sounds (ileus). Severe: hypoxia, hypovolaemic shock, oliguria. Grey Turner's (flanks) / Cullen's (periumbilical) = haemorrhage.	Almost all: abd pain . 50% as 'acute' episodes (each -> permanent damage); 35% relentless progressive pain; rest painless w/ diarrhoea. Pain: ↑ductal pressure + nerve involvement; eased leaning forward / alcohol. Weight loss, steatorrhea. DM 30% (70% if calcific). Erythema ab igne (hot water bottle).	Abd pain, weight loss, obstructive jaundice WITHOUT acute attacks.	Often asymptomatic until advanced: central abd pain, weight loss, obstructive jaundice . Pain = coeliac plexus invasion, gnawing, radiates to back, eased bending forward. Cachexia. 60% head of pancreas. Courvoisier : palpable GB + jaundice. May get diarrhoea, duodenal obstruction, DM, recurrent venous thrombosis.
Investigations	Dx : ↑ serum amylase or lipase + US/CT swelling . Lipase preferred (better accuracy). Amylase normalises 24-48h; persistently high = pseudocyst; massively ↑ peritoneal = ascites. Plain X-ray to exclude perforation/obstruction. US: gallstones, biliary obstruction, pseudocyst. CECT 6-10d for necrosis (↓enhancement); gas = infection.	Aims: confirm dx, define pancreatic function, show anatomy before surgery. (US/CT calcification, MRCP, faecal elastase, EUS – std).	↑ IgG / IgG4 + autoantibodies. Imaging: diffusely enlarged ('sausage') pancreas , narrowed pancreatic duct, lower bile duct stricture.	US + CECT . Staging: EUS / laparoscopy + lap US (size, vessels, mets). EUS/CT-guided biopsy if inoperable. MRCP/ERCP sensitive; ERCP main role = stent CBD for jaundice in inoperable.
Severity / Grading	CRP peak >210 mg/L in first 4d = severe (80% accuracy) . Serial CRP prognostic. Amylase has NO prognostic value . [Std scores: Glasgow-Imrie, Ranson, APACHE II; CT severity index/Balthazar].	— (graded by exocrine function loss; steatorrhea >90% destroyed).	—	TNM staging via EUS/laparoscopy. Resectable only 10-15% (most locally advanced).
Management	Steps: dx + severity; resuscitate (mild/severe); detect+treat complications; treat cause. Opiate analgesia : correct hypovolaemia w/ crystalloid (N saline). Severe -> HDU/ICU, CV line + urinary cath. O2; SIRS may need ventilation. Insulin for hyperglycaemia, IV Ca for hypocalcaemia. NG only if ileus. Early enteral feeding (NG = NJ; ↓endotoxaemia). LMWH prophylaxis. NO prophylactic abx ; infected necrosis -> carbapenem/quinolone + metronidazole. Cholangitis/jaundice + severe -> urgent ERCP . Gallstone (less severe): MRCP/EUS after acute phase; cholecystectomy within 2 wks of resolution (same admission).	1. Alcohol avoidance (halts progression, ↓pain). 2. Pain : NSAIDs; opiates (addiction risk); pregabalin/TCA low dose; pancreatic enzymes; coeliac plexus neurolysis (often relapses); total pancreatectomy last resort (causes brittle DM). 3. Malabsorption : low-fat diet (+MCT), pancreatic enzyme supplements, PPI (optimise duodenal pH). 4. Complications : pseudocyst, ascites, CBD/duodenal stricture, portal HTN.	Glucocorticoids (excellent response); some need azathioprine .	Surgical resection = only cure (5-yr ~12%); adjuvant gemcitabine -> 21-29%. Only 10-15% resectable. Most palliative. Palliative chemo FOLFIRINOX (5-FU, leucovorin, irinotecan, oxaliplatin) -> median 11 mo. Pain: analgesia / coeliac plexus neurolysis. Jaundice: choledochojejunostomy (fit) or stent (elderly/advanced). Ampullary/periampullary AC: less aggressive, 25% 5-yr after resection.
Complications	Pancreatic fluid collection -> pseudocyst (matures ~6 wks; >6 cm rarely resolves, may erode vessels -> pseudoaneurysm, palpable mass). Pancreatic ascites (duct leak). Pleural effusion / pleuro-pancreatic fistula. Necrosis, abscess, multi-organ failure. Pseudocyst drainage (cystogastrostomy) after >=6 wks; infected necrosis/abscess -> endoscopic drainage or MIRP necrosectomy.	DM, malabsorption/steatorrhea, pseudocyst, ascites, CBD/duodenal stricture, portal HTN, opiate dependence.	Can mimic/be mistaken for cancer.	Mets (nodes, liver), biliary/duodenal obstruction, recurrent venous thrombosis, acute pancreatitis.